

CPU & INSTRUCTION FORMATS PRACTICE SHEET

NIMCET TARGET PAPER • COMPUTER BASICS

ResourceOS

1. A CPU has a 16-bit instruction format. An instruction uses 5 bits for opcode and specifies two register operands. How many general-purpose registers can this CPU support?

- (A) 16
- (B) 32
- (C) 64
- (D) 128

2. In a relative addressing mode instruction, the offset value is -20. If the Program Counter (PC) contains 500 when the branch instruction is fetched (2-byte instruction size), what is the target address?

- (A) 480
- (B) 482
- (C) 500
- (D) 502

3. A computer uses a 32-bit instruction size with 6 bits for opcode and 1 bit for direct/indirect addressing mode specification. If memory is byte-addressable, what is the maximum addressable memory size per instruction?

- (A) 16 MB
- (B) 32 MB
- (C) 64 MB
- (D) 128 MB

4. Which sequence of register transfers corresponds to the instruction fetch micro-operations in a basic computer?

- (A) $MAR \leftarrow PC$; $MBR \leftarrow M[MAR]$; $PC \leftarrow PC + 1$; $IR \leftarrow MBR$
- (B) $MBR \leftarrow PC$; $MAR \leftarrow M[MBR]$; $IR \leftarrow MAR$
- (C) $IR \leftarrow PC$; $MAR \leftarrow M[IR]$; $MBR \leftarrow MAR + 1$
- (D) $PC \leftarrow MAR$; $MBR \leftarrow M[PC]$; $IR \leftarrow MBR$

5. In Indexed Addressing Mode, the effective address of an operand is obtained by adding the displacement/address field content to which register?

- (A) Program Counter
- (B) Index Register
- (C) Memory Buffer Register
- (D) Stack Pointer

6. A CPU has 32 registers and uses a 32-bit fixed-length instruction format. A 3-operand register instruction uses 6 bits for opcode. How many bits are unused in this instruction format?

- (A) 9 bits
- (B) 11 bits
- (C) 15 bits
- (D) 18 bits

7. What is the effective address in Indirect Addressing Mode if the instruction address field contains 2000, and memory location 2000 contains value 4500?

- (A) 2000
- (B) 4500
- (C) 6500
- (D) 2500

8. Which addressing mode allows program code relocatability in memory without modifying individual target addresses?

- (A) Direct Addressing Mode
- (B) PC-Relative Addressing Mode
- (C) Immediate Addressing Mode
- (D) Absolute Addressing Mode

9. In an expanding opcode scheme, a 16-bit instruction format supports both 2-address and 1-address instructions. If there are 14 two-address instructions (using 4-bit register fields), how many one-address instructions can be supported?

- (A) 16
- (B) 32
- (C) 2
- (D) 64

10. Which register holds the status flags resulting from the execution of arithmetic and logic operations in the ALU?

- (A) Instruction Register
- (B) Program Status Word (PSW)
- (C) Memory Address Register
- (D) Base Register

11. What main advantage does Register Indirect Addressing offer over Direct Addressing?

- (A) Eliminates memory accesses during execution
- (B) Requires fewer instruction bits to specify a large memory address
- (C) Stores operand values directly inside instruction code
- (D) Avoids using CPU general registers

12. In an accumulator-based architecture, how many explicit memory/register operand fields are required in arithmetic instructions?

- (A) Zero
- (B) One
- (C) Two
- (D) Three

13. A processor has 128 distinct instructions and 64 general registers. How many bits are required for a 2-operand instruction (Register-to-Register)?

- (A) 15 bits
- (B) 17 bits
- (C) 19 bits
- (D) 21 bits

14. Which addressing mode is most suitable for stepping through array elements stored sequentially in memory?

- (A) Immediate Addressing
- (B) Auto-increment / Indexed Addressing
- (C) Direct Addressing
- (D) Register Addressing

15. What is the role of the Microprogram Counter (μ PC) in a microprogrammed control unit?

- (A) Stores the main memory address of instructions
- (B) Holds the address of the next microinstruction in Control Memory
- (C) Counts the total number of CPU clock cycles
- (D) Decodes macro-instruction operation codes

16. If an instruction format allows 4-bit register field, 6-bit opcode, and 14-bit memory address field, what is the minimum total instruction length?

- (A) 20 bits
- (B) 24 bits
- (C) 28 bits
- (D) 32 bits

17. During a CALL subroutine instruction, what register value is saved onto the CPU system stack?

- (A) Accumulator (AC)
- (B) Return Address stored in Program Counter (PC)
- (C) Instruction Register (IR)
- (D) Memory Address Register (MAR)

18. What is the key operational feature of a Hardwired Control Unit compared to a Microprogrammed Control Unit?

- (A) Slower speed and flexible reprogramming
- (B) Faster execution speed due to combinational logic circuit implementation
- (C) Uses Control ROM memory for microcode instructions
- (D) Simplifies complex instruction set additions

19. In Base Register Addressing, how is the physical effective address calculated?

- (A) $EA = [Base\ Register] + Offset$ (B) $EA = [Program\ Counter] + Offset$
(C) $EA = [Base\ Register] \times Offset$ (D) $EA = M[[Base\ Register] + Offset]$

20. In a zero-address stack architecture, how is the expression $(A + B) * C$ evaluated using Reverse Polish Notation (RPN)?

- (A) PUSH A, PUSH B, ADD, PUSH C, MUL (B) PUSH A, ADD B, MUL C
(C) ADD A B, MUL C (D) PUSH A, PUSH B, PUSH C, ADD, MUL

21. How many total memory reference accesses are required to fetch and execute an instruction using Memory Indirect Addressing Mode?

- (A) 1 access (B) 2 accesses
(C) 3 accesses (D) 0 accesses

22. Which bus lines carry interrupt request signals from peripheral controllers to the CPU?

- (A) Data Bus (B) Control Bus
(C) Address Bus (D) System Multiplexer

23. In an instruction code, if 4 bits are reserved for mode selection, how many distinct addressing modes can be specified?

- (A) 8 (B) 16
(C) 32 (D) 4

24. What is the primary difference between a Branch instruction and a Call Subroutine instruction?

- (A) Branch changes PC; Call changes PC and saves return address (B) Branch saves return address; Call does not address
(C) Call requires direct addressing mode only (D) Branch requires an explicit stack pointer

25. In a 16-bit processor, if an immediate operand instruction is `MOV R1, #00FFh`, where is the constant value 00FFh fetched from during processing?

- (A) CPU Accumulator register (B) Memory immediately following opcode in instruction fetch phase
(C) System stack top (D) Data cache tag directory

ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(B)	6	(B)	11	(B)	16	(B)	21	(C)
2	(B)	7	(B)	12	(B)	17	(B)	22	(B)
3	(B)	8	(B)	13	(C)	18	(B)	23	(B)
4	(A)	9	(B)	14	(B)	19	(A)	24	(A)
5	(B)	10	(B)	15	(B)	20	(A)	25	(B)